

Perspective-n-Point Problem

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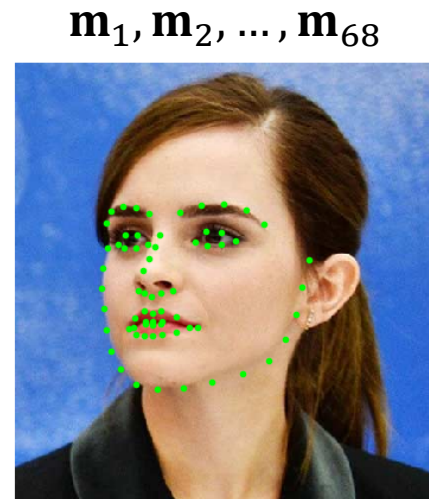
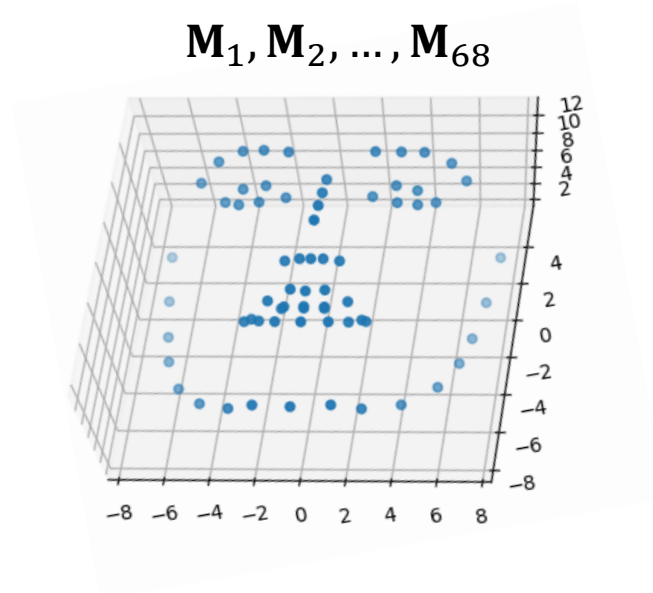
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Definition of PnP

- Given $\mathbf{K}$, $\mathbf{m}_i$, $\mathbf{M}_i$ for $i = 1, 2, \dots, n$
 - $\mathbf{K}$ is the intrinsic matrix of shape (3,3)
 - $\mathbf{m}_i$ is the i th detected 2D landmark of shape (2,1)
 - $\mathbf{M}_i$ is the i th 3D landmark of shape (3,1)
- Find $(\mathbf{R}, \mathbf{t})$ such that
 - $\tilde{\mathbf{m}}_i = \mathbf{K}[\mathbf{R} \ \mathbf{t}]\tilde{\mathbf{M}}_i$ with $\tilde{\mathbf{M}}_i = \begin{bmatrix} \mathbf{M}_i \\ 1 \end{bmatrix}$, $\tilde{\mathbf{m}}_i = \lambda_i \begin{bmatrix} \mathbf{m}_i \\ 1 \end{bmatrix}$ for $i = 1, 2, \dots, n$
 - $\mathbf{R}$ is a rotation matrix, i.e. $\mathbf{R}^T \mathbf{R} = \mathbf{I}$.
 - $\mathbf{R}$ is of shape (3,3), $\mathbf{t}$ is of shape (3,1)

Head Pose Estimation

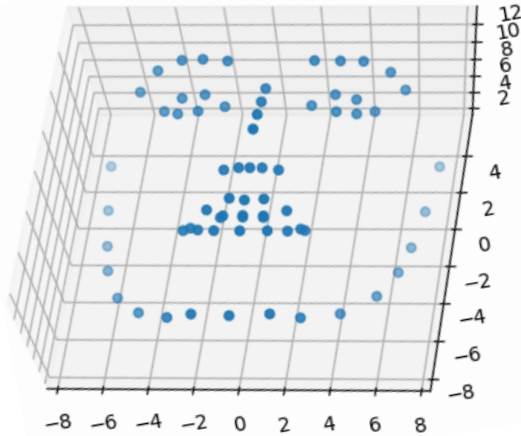


Given $\mathbf{K}$, $\mathbf{m}_i$, $\mathbf{M}_i$ for $i = 1, 2, \dots, 68$

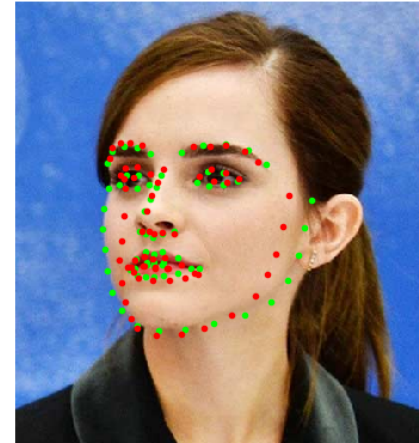
- $\mathbf{K}$ is the intrinsic matrix of shape (3,3)
- $\mathbf{M}_i$ is the i th 3D landmark of shape (3,1)
- $\mathbf{m}_i$ is the i th detected 2D landmark of shape (2,1)

Head Pose Estimation (cont.)

$\mathbf{M}_1, \mathbf{M}_2, \dots, \mathbf{M}_{68}$



$\mathbf{m}_1, \mathbf{m}_2, \dots, \mathbf{m}_{68}$



Find $(\mathbf{R}, \mathbf{t})$ such that

- $\tilde{\mathbf{m}}_i \sim \mathbf{K}[\mathbf{R} \ \mathbf{t}]\tilde{\mathbf{M}}_i$ with $\tilde{\mathbf{M}}_i = \begin{bmatrix} \mathbf{M}_i \\ 1 \end{bmatrix}$, $\tilde{\mathbf{m}}_i = \lambda_i \begin{bmatrix} \mathbf{m}_i \\ 1 \end{bmatrix}$ for $i = 1, 2, \dots, 68$
- $\mathbf{R}$ is a rotation matrix, i.e. $\mathbf{R}^T \mathbf{R} = \mathbf{I}$.
- The green points are detected facial landmarks, while the red points are the re-projected facial landmarks.
- Re-projection error refers to the difference between the green and red points.

Remarks

- Find $(\mathbf{R}, \mathbf{t})$ such that
 - $\tilde{\mathbf{m}}_i = \mathbf{K}[\mathbf{R} \ \mathbf{t}] \tilde{\mathbf{M}}_i$ with $\tilde{\mathbf{M}}_i = \begin{bmatrix} \mathbf{M}_i \\ 1 \end{bmatrix}$, $\tilde{\mathbf{m}}_i = \lambda_i \begin{bmatrix} \mathbf{m}_i \\ 1 \end{bmatrix}$ for $i = 1, 2, \dots, n$
 - $\mathbf{R}$ is a rotation matrix, i.e. $\mathbf{R}^T \mathbf{R} = \mathbf{I}$.
 - $\mathbf{R}$ is of shape (3,3), $\mathbf{t}$ is of shape (3,1)
- Questions
 - Is this problem linear or nonlinear?
 - How many variables are there to be solved?

Solve the Equation

Given $\mathbf{K}$, (x_i, y_i) , (X_i, Y_i, Z_i) for $i = 1, 2, \dots, n$.

Solve $(\mathbf{R}, \mathbf{t})$ such that

$$\tilde{\mathbf{m}}_i = \mathbf{K}[\mathbf{R} \ \mathbf{t}]\tilde{\mathbf{M}}_i$$

$$\mathbf{K}^{-1}\tilde{\mathbf{m}}_i = [\mathbf{R} \ \mathbf{t}]\tilde{\mathbf{M}}_i$$

$$\mathbf{K}^{-1} \begin{bmatrix} w_i x_i \\ w_i y_i \\ w_i \end{bmatrix} = \begin{bmatrix} \lambda_i u_i \\ \lambda_i v_i \\ \lambda_i \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \begin{bmatrix} X_i \\ Y_i \\ Z_i \\ 1 \end{bmatrix}$$

Solve the Equation (cont.)

$$\begin{bmatrix} \lambda_i u_i \\ \lambda_i v_i \\ \lambda_i \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \begin{bmatrix} X_i \\ Y_i \\ Z_i \\ 1 \end{bmatrix}$$

$$\Rightarrow \begin{cases} \lambda_i u_i = r_{11}X_i + r_{12}Y_i + r_{13}Z_i + t_1 \\ \lambda_i v_i = r_{21}X_i + r_{22}Y_i + r_{23}Z_i + t_2 \\ \lambda_i = r_{31}X_i + r_{32}Y_i + r_{33}Z_i + t_3 \end{cases}$$

$$\Rightarrow \begin{cases} r_{11}X_i + r_{12}Y_i + r_{13}Z_i + t_1 - r_{31}X_i u_i - r_{32}Y_i u_i - r_{33}Z_i u_i - t_3 u_i = 0 \\ r_{21}X_i + r_{22}Y_i + r_{23}Z_i + t_2 - r_{31}X_i v_i - r_{32}Y_i v_i - r_{33}Z_i v_i - t_3 v_i = 0 \end{cases}$$

Solve the Equation (cont.)

$$\begin{cases} r_{11}X_i + r_{12}Y_i + r_{13}Z_i + t_1 - r_{31}X_iu_i - r_{32}Y_iu_i - r_{33}Z_iu_i - t_3u_i = 0 \\ r_{21}X_i + r_{22}Y_i + r_{23}Z_i + t_2 - r_{31}X_iv_i - r_{32}Y_iv_i - r_{33}Z_iv_i - t_3v_i = 0 \end{cases}$$

$$\begin{array}{l} \text{Point 1} \\ \text{Point 2} \\ \text{Point 3} \\ \vdots \end{array} \begin{bmatrix} X_1 & Y_1 & Z_1 & 0 & 0 & 0 & -X_1u_1 & -Y_1u_1 & -Z_1u_1 & 1 & 0 & -u_1 \\ 0 & 0 & 0 & X_1 & Y_1 & Z_1 & -X_1v_1 & -Y_1v_1 & -Z_1v_1 & 0 & 1 & -v_1 \\ X_2 & Y_2 & Z_2 & 0 & 0 & 0 & -X_2u_2 & -Y_2u_2 & -Z_2u_2 & 1 & 0 & -u_2 \\ 0 & 0 & 0 & X_2 & Y_2 & Z_2 & -X_2v_2 & -Y_2v_2 & -Z_2v_2 & 0 & 1 & -v_2 \\ \vdots & & & & & & & & & & & \end{bmatrix} \begin{bmatrix} r_{11} \\ r_{12} \\ r_{13} \\ r_{21} \\ r_{22} \\ r_{23} \\ \vdots \\ t_1 \\ t_2 \\ t_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

- Recall that r_{ij} is the entry of a rotation matrix, and r_{ij} needs to satisfy $\mathbf{R}^T \mathbf{R} = \mathbf{I}$.

Different Approaches

- CV_ITERATIVE
 - Iterative methods based on Levenberg-Marquardt optimization.
- CV_P3P
 - Complete Solution Classification for the Perspective-Three-Point Problem, PAMI, 2003.
- CV_DLS
 - A Direct Least-Squares (DLS) Method for PnP, ICCV, 2011.
- CV_EPNP
 - EPnP: Efficient Perspective-n-Point Camera Pose Estimation, IJCV, 2009.
- CV_SQPNP
 - A Consistently Fast and Globally Optimal Solution to the Perspective-n-Point Problem, ECCV, 2020.

SolvePnP() in OpenCV

```
ret, rvec, tvec = cv2.solvePnP(3d_pts, 2d_pts, K, disc, flags)
```

- K is the intrinsic matrix.
- **disc** is the distortion coefficients.
- **rvec** is the rotation vector. Use `cv2.Rodrigues()` to convert it to a rotation matrix.
- **flags** can be

```
cv2.SOLVEPNP_ITERATIVE = 0, (Levenberg–Marquardt optimization)  
cv2.SOLVEPNP_EPNP = 1,  
cv2.SOLVEPNP_P3P = 2,  
cv2.SOLVEPNP_DLS = 3,  
cv2.SOLVEPNP_SQPNP = 8
```

cv2.Rodrigues

- Converts a rotation matrix to a rotation vector of vice versa.
- A rotation vector is a convenient and compact representation of a rotation matrix.

```
R, _ = cv2.Rodrigues(rvec)  
rvec, _ = cv2.Rodrigues(R)
```

$$\theta \leftarrow \text{norm}(\mathbf{r})$$

$$\mathbf{r} \leftarrow \mathbf{r}/\theta$$

$$\mathbf{R} = \cos(\theta) \mathbf{I} + (1 - \cos(\theta)) \mathbf{r} \mathbf{r}^T + \sin(\theta) \begin{bmatrix} 0 & -r_z & r_y \\ r_z & 0 & -r_x \\ -r_y & r_x & 0 \end{bmatrix}$$